

FitBuddy AI — Coding & Solution

Phase 5 • Project Development Phase

Implementation summary

- The documented implementation uses Python/FastAPI for the web server and REST API, Pydantic for request/response validation, SQLite for local persistence, HTML/CSS/JavaScript for the frontend, optional Gemini integration for plan generation, and pytest for automated testing.
- The project can run without a live Gemini key when demo mode is enabled.

Core API implementation

- POST /api/users — create/update profile.
- POST /api/users/{user_id}/plans/generate — generate plan.
- POST /api/users/{user_id}/feedback — save feedback and update plan.
- GET /api/users — list users.
- GET /api/coach/summary — coach summary.
- GET /api/feedback?limit=20 — recent feedback.
- GET /health — health check.
- GET /docs — Swagger/OpenAPI docs.

Generation behavior

- The project documents a deterministic local generator for demo mode.
- When live Gemini generation is enabled, the project is intended to use Gemini for plan generation and fall back to a local demo plan if the request fails.